TCS Interview: Others

1. ACID Properties (DB — important)

👉 Properties that ensure **reliable transactions in a database**

A – Atomicity

👉 “All or nothing”

- If one part fails → whole transaction fails
- Example: money transfer

C – Consistency

👉 Database stays valid

- Rules/constraints maintained
- No invalid data

I – Isolation

👉 Transactions don’t interfere

- Multiple users → no conflict

D – Durability

👉 Once committed → permanent

- Even if system crashes

Interview answer

“ACID properties ensure reliable and consistent database transactions by enforcing atomicity, consistency, isolation, and durability.”

2. FastAPI (just basics)

What is FastAPI?

👉 Python framework for building APIs

Answer

“FastAPI is a modern Python web framework used to build fast and efficient APIs with automatic validation and high performance.”

Why used?

- Fast (async support)
- Easy to use
- Automatic docs

3. Synchronization (important for threading)

What is Synchronization?

👉 Controlling access to shared resources

Why needed?

If multiple threads access same data:

👉 leads to **race condition**

Answer

“Synchronization ensures that multiple threads access shared resources in a controlled manner to avoid inconsistencies.”

4. Multithreading vs Async (light level)

Multithreading

👉 Multiple threads run simultaneously

Used for: parallel tasks

Async

👉 non-blocking execution

Used for: I/O tasks (API calls, DB)

Simple difference

Multithreading	Async
parallel	non-blocking
CPU tasks	I/O tasks

Think of a restaurant. A synchronous waiter takes an order, waits for the kitchen to cook it (does nothing else), serves it, and *then* takes the next table's order. An asynchronous waiter takes an order, sends it to the kitchen, and immediately takes the next table's order while the first meal cooks

5. Static & Final (basic idea)

👉 From Java, but conceptually useful

Static

👉 Belongs to class, not object

The static keyword indicates that a member belongs to the **class itself** rather than to any specific instance (object).

Final

👉 Cannot be changed

The final keyword is used to restrict the user from changing or overriding an entity.

🧠 Answer

“Static members belong to the class, while final variables cannot be modified once assigned.”

6. MS Word basics (don’t overthink)

They may ask:

- bold / italic
- alignment
- save / save as
- shortcuts

👉 Keep it basic.

- **Ctrl + N**: Create a new document.
- **Ctrl + S**: Save the current document.
- **Ctrl + K**: Insert a hyperlink.
- **Ctrl + Enter**: Insert a page break.
- **Ctrl + F / H**: Find text / Find and Replace.
- **Ctrl + E**: Center align text.
- **Alt + Shift + D**: Insert the current date.